

B/S Miller
Rishu
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Acid

Aliah University

End-Semester Examination (Autumn Semester) - 2023

(For 1st Year 1st Semester MCA)

Paper Name: Database Management Systems

Paper Code: MCAPGPC02

Full Marks: 80

Time: 3 hrs

Group-A

Answer all the following questions.

5 × 2 = 10

1. [Redacted]

Group-B

Answer any five questions.

5 × 6 = 30

2. [Redacted] 6
3. What is DDL? Define procedural and non procedural DML. 2+4=6
4. What is Normalization? Explain 1NF and 2NF with suitable examples. 2+4=6
5. What do you mean by shared and exclusive lock? Discuss 2PL. 4+2=6
6. BCNF is stronger than 3NF. Justify your answer. 6
7. Consider the relation schemas R(A,B,C,D,E,F,G) with the following functional dependency: AB → CD, D → E, C → FG. Find all the candidate keys of R. 6

Group-C

Answer any four questions.

4 × 10 = 40

8. Create the table Student (Rollno, Name, CourseId, emailId, phoneno) and insert data (4 rows) into the table.

Write the following SQL queries:

- Find out list of students who have enrolled in "computer" course.
- Add a column DoB in student table.

iii) List name of students having character 'b' in their name. $4 + (2 \times 3) = 10$

9. Define Transaction. State the ACID properties of transaction and explain its state diagram. $2+4+4=10$

10. Explain FD, Trivial and non Trivial Dependency. State Armstrong's Axioms. $5+5=10$

11. Consider the following Relation R(A,B,C) is decomposed into R1(A,B) AND R2(B,C).

A	B	C
a	b	a
b	b	b
c	c	a

Check the decomposition is lossless or lossy. 10

12. What is conflict and view serializability? Check the following schedule is Conflict serializable or not. 10

T1	T2	T3
	Read(z)	
	Read(y)	
	Write(y)	
		Read(y)
		Read(y)
Read(x)		
Read(x)		
		Write(y)
		Write(z)
	Read(x)	
Read(y)		
Write(y)		
	Read(y)	
	Write(y)	
	Read(x)	
	Write(x)	

Q. 2
Discrete Mathematics
Revision

End - Semester Examination (Autumn Semester) 2022 - 23
(For 1st Year 1st Semester MCA)

AUCSE/044/2022 - 23

Paper Name: Computational Mathematics
Exam Code: MCABGPC04

Full Marks: 80
Time: 3 hrs

Answer any eight of the following.

[8 x 10 = 80]

1. a) What is the difference between interpolation and extrapolation? Give suitable examples. What is a polynomial?
b) The following table shows the viscosity of oil as a function of temperature. Find the viscosity of oil at a temperature of 140° using Lagrange interpolation formula.

Temp. °:	10	130	160	190
Viscosity :	10.8	8.1	5.2	4.8

2. a) Find the inverse of the following matrix by Gauss Elimination and Gauss Jordan method.

$$\begin{pmatrix} 3 & -1 & 1 \\ 1 & 2 & -3 \\ 5 & -2 & 2 \end{pmatrix}$$

(5 x 2 = 10)

3. Determine the value of $\sin 37^\circ$ by Newton's forward and backward difference interpolation formula.

(5 x 2 = 10)

x (degree):	15	20	25	30	35	40
f(x):	0.2588190	0.3420201	0.4226183	0.5	0.5735764	0.6427876

Round off to 3 places of decimal.

4. a) Find the root of the equation $f(x) = x^3 - 2x - 5$ in the interval (1, 2) by Newton Raphson and Secant method correct to 3 places of decimal. (5 + 5)
5. a) Deduce Simpson's 1/3rd rule from Newton's - Cote's quadrature formula. (4)
b) Integrate $\int_0^1 (x^3 - 2x^2) dx$, taking $h = 0.5$ and 0.25 , respectively using Romberg Integration. (6)
6. a) Calculate the approximate value of $\int_0^{\pi/2} \tan x dx$ by Composite Trapezoidal Rule by using 11 ordinates.
b) Calculate the approximate value of $\int_3^5 x dx$ with $h = 0.2$ using Simpson's 3/8th rule. (5 + 5)
7. What is a singular matrix? Find the trace, determinant and rank of the matrix in Question 2. (1 + 1 + 4)
8. Find the Adjoint of the matrix in Question 2 and hence find its inverse. Which of the matrices is orthogonal?
 $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$ (6 + 4)
9. Find all the measures of central tendency, range and interquartile range, variance and standard deviation for the following test scores recorded in a class: 88, 77, 95, 79, 59, 20, 86, 94, 68, 67, 91, 81, 72, 61, 60. (10)
10. a) A card is drawn from a full deck of 52 cards. What is the probability that the card drawn will be either a jack or a red card?
b) What is the probability that we roll a pair of six-sided fair dice and either get at least one 1, or an even sum when we add the outcomes of dice together?
c) State the Bayes' Theorem. If we draw a card from a deck of playing cards, and then draw a second card, what is the probability that we'll get two Aces in a row without replacing it. (3 + 3)

By Sonu Malviya
Performance
Portfolio

Aliah University
End-Semester Examination (Autumn Semester) - 2022
(For 1st Year 1st Semester MCA)

Full Marks: 80
Time: 3 hrs

Name: Computer Organization and Architecture
Paper Code: PGPC03

$5 \times 6 = 30$

(Answer any six questions)

1. Explain Von Neumann Bottleneck. How is this bottleneck addressed? 2+3
2. Describe the two different ways in which control units are designed. 2.5+2.5
3. Explain the role of an I/O interface and I/O controller. 2.5+2.5
4. Describe how the electronic computers have improved over the years with different generations of computers. 5
5. Draw a figure to explain how handshaking technique is used in asynchronous data transfer. 5
6. Write a short note on magnetic disk with a diagram. 5
7. A hierarchical memory system has the following specifications-
i) Cache access time of 30 sec (ii) Memory access time of 200 sec (iii) 78% of memory requests are for read (iv) Hit ratio of 0.92 for read access and miss through scheme is used. Calculate the following:
(a) Average access time of the memory system considering only read cycle and 2.5+2.5
(b) Average access time of the system both for read and write requests.

$10 \times 5 = 50$

Group-B (Answer any five questions)

1. Draw and describe the memory hierarchy of a computer system in terms of performance and cost. 5
2. Describe with example the different types of computers based on performance, size and cost. 5
3. What are the two dimensions of locality of reference? What do you mean by write through and write back policies? Describe the different cache mapping techniques. 2+2+6
4. Explain the three different cache categories. Now explain what are the three techniques to reduce it. 10
5. Describe in detail the different modes of data transfer between I/O devices and the central computer. 5+5
6. Differentiate Sequential, Random, Set Associative and Associative accessing methods. Write the names of Flynn's machine organizations. 8+2
7. Give any six major characteristics each of RISC and CISC. Point out any two merits of both RISC and CISC. Give an example of each of these. 6+2+2

Arshad Zahid Rahmon
Rahmon

Aliah University

Odd Semester Examination (Autumn Semester 2022-23)

1st year MCA

Paper Code: MCAPGPC01

Paper Name: Discrete Mathematics and Graph theory

Full marks: 80

Time : 3 hours

Group A

Answer all the questions

5X4=20

1. Define planar and non planar graph. How many vertices does a regular graph have if its order is 4 and there are 10 edges?
2. In how many ways can we select three students from a group of five students to stand in line for a picture?
3. Define eccentricity of a vertex. Differentiate vertex disjoint and edge disjoint subgraphs.
4. How many ways are there to select five players from a 10-member team to make a trip to a match at another school?
5. Define walk, path and circuit.

Group B

Answer any 6 questions

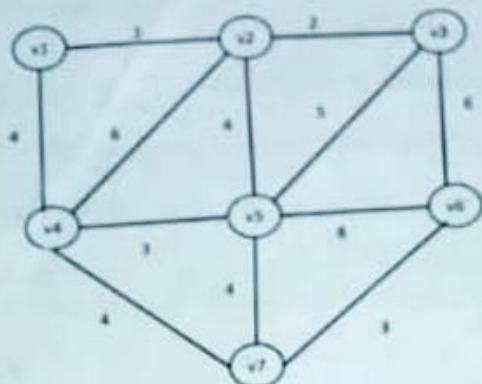
6X10=60

1. a) A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, 1 in Java and Python, 5 in C# and Python, and 1 programmer is proficient in all three languages above. Determine the number of computer programmers that are not proficient in any of these three languages. (6)
b) Among 50 patients admitted to a hospital, 25 are diagnosed with pneumonia, 30 with bronchitis, and 10 with both pneumonia and bronchitis. Determine:
(i) The number of patients diagnosed with pneumonia or bronchitis (or both).
(ii) The number of patients not diagnosed with pneumonia or bronchitis. (4)
2. a) State and prove pigeonhole principle. (4)
b) Find the minimum number of students should be there in a class so that 3 students have same birth month. (3)
c) There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, what is the minimum number of different rooms that will be needed? (3)
3. a) Solve the HDE

$$y_{n+2} - 2y_{n+1} - 3y_n = 0$$
$$y_0 = 1, y_1 = 0$$

b) Find the generating function of the infinite sequence $1, k, k^2, k^3, \dots$ where k is a fixed constant. (5+5)

4) Define spanning tree and minimum spanning tree. Using Prim's algorithm, calculate the minimum spanning tree for the given graph.



$$\begin{array}{r} 120 \\ 56 \\ \hline 64 \end{array}$$

$$\begin{array}{r} 120 \\ 56 \\ \hline 64 \\ 64 \\ \hline 128 \\ 128 \\ \hline 256 \\ 256 \\ \hline 512 \\ 512 \\ \hline 1024 \\ 1024 \\ \hline 2048 \\ 2048 \\ \hline 4096 \\ 4096 \\ \hline 8192 \\ 8192 \\ \hline 16384 \\ 16384 \\ \hline 32768 \\ 32768 \\ \hline 65536 \\ 65536 \\ \hline 131072 \\ 131072 \\ \hline 262144 \\ 262144 \\ \hline 524288 \\ 524288 \\ \hline 1048576 \\ 1048576 \\ \hline 2097152 \\ 2097152 \\ \hline 4194304 \\ 4194304 \\ \hline 8388608 \\ 8388608 \\ \hline 16777216 \\ 16777216 \\ \hline 33554432 \\ 33554432 \\ \hline 67108864 \\ 67108864 \\ \hline 134217728 \\ 134217728 \\ \hline 268435456 \\ 268435456 \\ \hline 536870912 \\ 536870912 \\ \hline 1073741824 \\ 1073741824 \\ \hline 2147483648 \\ 2147483648 \\ \hline 4294967296 \\ 4294967296 \\ \hline 8589934592 \\ 8589934592 \\ \hline 17179869184 \\ 17179869184 \\ \hline 34359738368 \\ 34359738368 \\ \hline 68719476736 \\ 68719476736 \\ \hline 137438953472 \\ 137438953472 \\ \hline 274877906944 \\ 274877906944 \\ 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